R6.8 Transport oracles on factor-base points (Proposition 5′)

Round-6 lead R6.8. Toy instances (p ≈ 2^18–2^20); no sub-rho claim; no deployed-curve claim. Full proof and tables: `RESULTS.md`.

Result

Proposition 5′ (proved). An index-calculus-style attacker with transport oracles T_ψ on arbitrary formed points is simulated exactly, with the same success probability and no dearer decomposition calls, by an attacker on E_1 without oracles whose factor base is the union of pulled-back bases $F \cup \bigcup_\psi \hat{\psi}(F'_\psi)$, of size at most $(1 + |\Psi|)|F|$ and independent of the instance. Pullbacks only grow (Thm 4.2, Lemma 5); the degree enters through the fibre size $|M_k| \le \deg \psi$ and the image index $i \le \min(\deg \psi, h)$. No exponent moves.

Falsifier (measured, exact census): did not fire in 0 of 16 cells (4 curves × |F| ∈ {72, 144} × m ∈ {2, 3}). Union class factor per unknown / best single base: 0.02–0.27, threshold > |Ψ| ∈ {1, 2}.

Measured gains are oracle-free: pulled-back bases gain the image index (≈3× on j = 0 with ℓ = 3, ≈2× for ℓ = 2 on the 6r curve), matched within 2% by filtering small-x points into ℓE (no oracle). At m = 2 the union costs 0.31–0.44 of plain (derived, §15.2 model), inside the predicted interval [1/(k·i), 1/k]. m = 3 at |F| = 144 is near saturation (D ≈ r), a toy-size effect.

Verdict: confirmed. No §15.4 row moves.

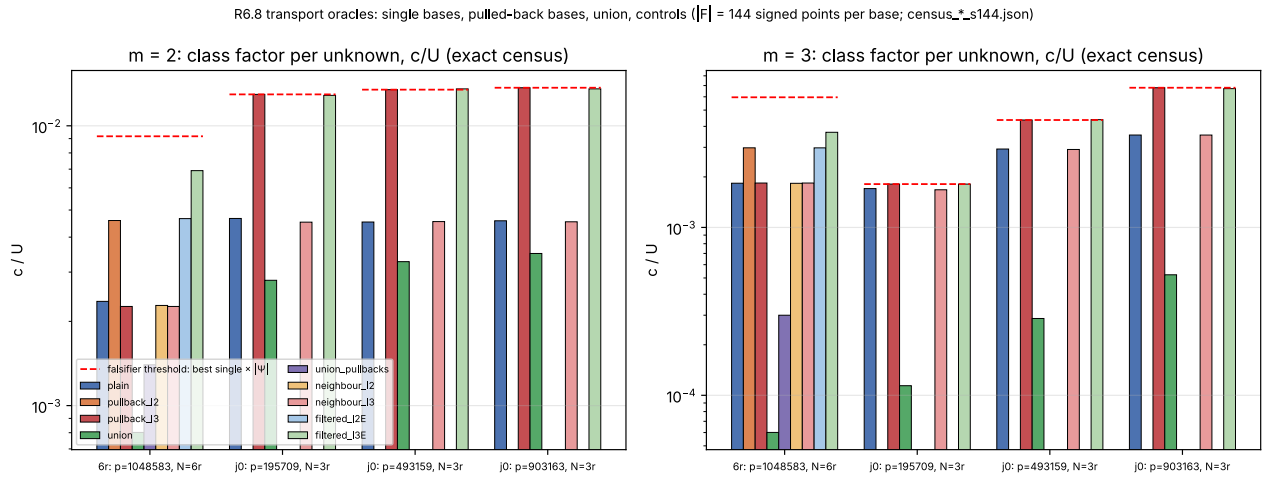


Figure 1: Class factor per unknown by arm (exact census, |F| = 144); red dashed: falsifier threshold best single × |Ψ|.

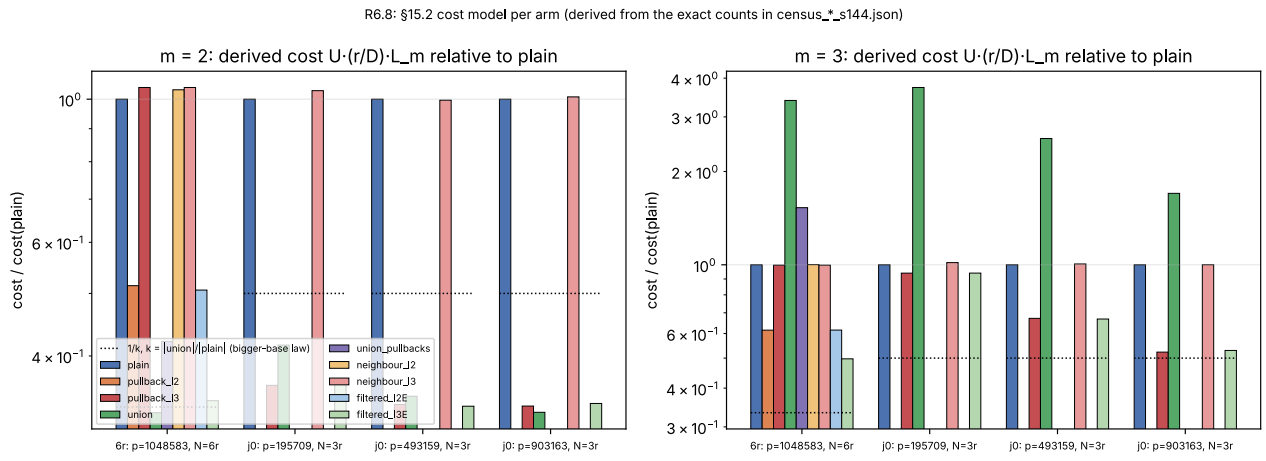


Figure 2: Derived §15.2 cost relative to the plain base; dotted: 1/k.

Limits

Four toy curves, one seed; class-factor bound (5.2) heuristic; non-IC attackers, transport of non-formed points and the CJ-DLP side are out of scope.